The impact of weather patterns on annual fluctuations in tea yield in northern Iran: A machine learning approach (1961-2023)

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Highlights

- Tea yield variability in northern Iran is analyzed using ERA5 reanalysis data
- A Weather Pattern Index (WP Index) is developed to quantify synoptic influences
- Machine learning models are applied to predict yield from atmospheric conditions
- Frequent patterns such as WP18 and WP1 are linked to favorable tea yield outcomes
- The framework supports climate-resilient decision-making in tea-producing regions

Abstract: Interannual variability in crop yields remains a key barrier to sustainable agriculture, particularly under the intensifying influence of atmospheric dynamics and climate change. This study explores how large-scale weather patterns (WPs) affect annual fluctuations in tea yields across Iran's Gilan and Mazandaran provinces from 1961 to 2023. Utilizing ERA5 reanalysis data, we identified 30 distinct WPs using the MiniBatch K-Means clustering algorithm, informed by synoptic classification schemes such as MO30. Annual yield years were stratified into high-, medium-, and low-yield groups. WP1, WP18, and WP9 were consistently associated with above-average yields, while WP24, WP12, and WP0 correlated with significant reductions. We developed a Weather Pattern Index (WP Index) based on the frequency contrast between favorable and unfavorable WPs, which accounted for a substantial portion of interannual yield variance. Furthermore, machine learning models—including linear regression, support vector regression (SVR), and Random Forest—showed high predictive performance despite the limited data. Among all climatic predictors, near-surface air temperature (t2m) and dew point temperature (d2m)

emerged as dominant drivers. The proposed framework offers a scalable and actionable tool for climate-resilient agricultural planning and adaptive yield forecasting in tea-producing regions.

Keywords: Machine learning, Tea performance, Weather patterns, Climate change, Northern Iran, Agricultural forecast

INTRODUCTION

Agriculture is one of the major pillars of sustainable development, food security, and economic growth in developing countries—such as Iran—where most of the population depends on livelihoods associated with natural resources and agriculture [1, 2]. Not only does the sector supply food and raw materials to agro-industries but also contributes to employment generation, poverty alleviation, rural population maintenance, and rural-to-urban migration prevention. Within this broader perspective, certain high-value crops have an disproportionate role in the sustenance of local economies and the ecosystem. Of these, tea strategically occupies northern Iran, especially in the provinces of Gilan and Mazandaran. The traditional crop has remained at the center of agricultural production in the region. As of 2023, roughly 14,357 hectares of lands in Mazandaran and Gilan are under tea cultivation [11]. Tea production is a significant source of livelihood for many rural households and contributes to the local economy significantly through value-added chains including processing, packing, and marketing. Moreover, tea cultivation on slopes is liable for soil stabilization, reducing the threat of erosion, conserving agricultural land cover, and ensuring biodiversity conservation in the humid subtropical ecosystem in northern Iran [3].

Tea production in northern Iran—particularly in the humid, subtropical provinces of Gilan and Mazandaran is susceptible to inter-annual climate variability. In this region, tea plants are primarily cultivated on sloped terrains that depend on consistent rainfall, moderate temperatures, and stable subsoil moisture during the growing season (April to October). Variations in seasonal precipitation, increases in average temperatures, and reductions in mid- and deep-layer soil moisture can significantly disrupt physiological processes such as bud formation, photosynthesis, and leaf expansion. These climatic stresses often lead to yield reductions and deterioration in leaf quality . In addition to individual climatic variables, recent research highlights the critical role of large-scale atmospheric circulation patterns—called Weather Patterns (WPs)—in shaping regional climatic conditions. Persistent or recurring WPs can intensify drought spells, delay rainfall onset, or concentrate precipitation in short bursts, influencing the water and thermal conditions essential for tea plant development. For instance, dry anticyclonic WPs can suppress rainfall and increase evapotranspiration, aggravating water stress in the crop, whereas humid cyclonic WPs may enhance cloud cover and soil moisture, promoting growth. The frequency, duration, and timing of these WPs during the tea growing season are thus key determinants of yield quantity and quality[16]. This complex interaction between weather patterns and crop response poses serious challenges for short-term management and long-term adaptation strategies in Iran's tea-producing regions.

Under such circumstances, a precise and data-driven understanding of climate and tea yield relationships—primarily through statistical modeling and machine learning—can pave the way for developing early warning tools and designing climate adaptation policies in the agricultural sector. Recent advances in deep learning and computer vision have shown promise in addressing complex detection challenges of farm environments, including crop residue mapping and field monitoring under noisy conditions [5]. These developments support the broader integration of AI-based tools in agrometeorological research and precision farming. In recent years, extreme climate events' increasing intensity and frequency have underscored the need for a deeper understanding of their impacts on agricultural systems—especially in climate-sensitive and dependent regions such as northern Iran.

In this context, the use of weather patterns as tools to describe the large-scale structure of the climate has gained growing traction in agrometeorological studies [4]. Unlike traditional analyses focusing on individual climate variables such as average temperature or total precipitation, the Weather Pattern (WP) approach emphasizes the simultaneous spatial configuration of multiple climate variables, providing a more accurate representation of the persistent conditions affecting plant growth [6]. Globally, the application of weather pattern classifications in agricultural research is expanding. For example, in the United Kingdom, the UK Met Office introduced the MO30 classification system, offering a framework to assess the impacts of WPs on agriculture, water resources, and ecosystems. Building on this innovation, numerous studies in Europe and North America have used reanalysis data to extract weather patterns and analyze their links to crop yields. However, such approaches remain underexplored in the Middle East, particularly in Iran, where most agricultural studies still rely on individual climate indicators rather than integrated pattern-based frameworks [7].

While weather pattern (WP) frameworks—such as the MO30 system—have been widely applied in European contexts (Philipp et al., 2007), no similar studies have yet explored their relevance to Iran's agroclimatic zones, despite their potential value for climate-resilient agriculture.

Moreover, recent work by Sahu et al. (2024) on Darjeeling tea in India highlights the critical role of synoptic-scale weather variability in perennial crop yields, emphasizing the broader applicability of WP-based analysis in tea-producing regions.

Therefore, this study is designed to fill this gap by combining weather pattern classification, long-term time series analysis of tea yield, and applying statistical and machine learning techniques [8]. This integrated approach aims to quantitatively and qualitatively assess the structural relationships between atmospheric circulation regimes and tea productivity in northern Iran. The study also introduces a scalable framework for analyzing climate-related risks in other mountainous and humid regions with perennial crops by addressing a critical knowledge gap. (Fig 1) provides a visual representation of the MO30 and offers a detailed description of each WP.

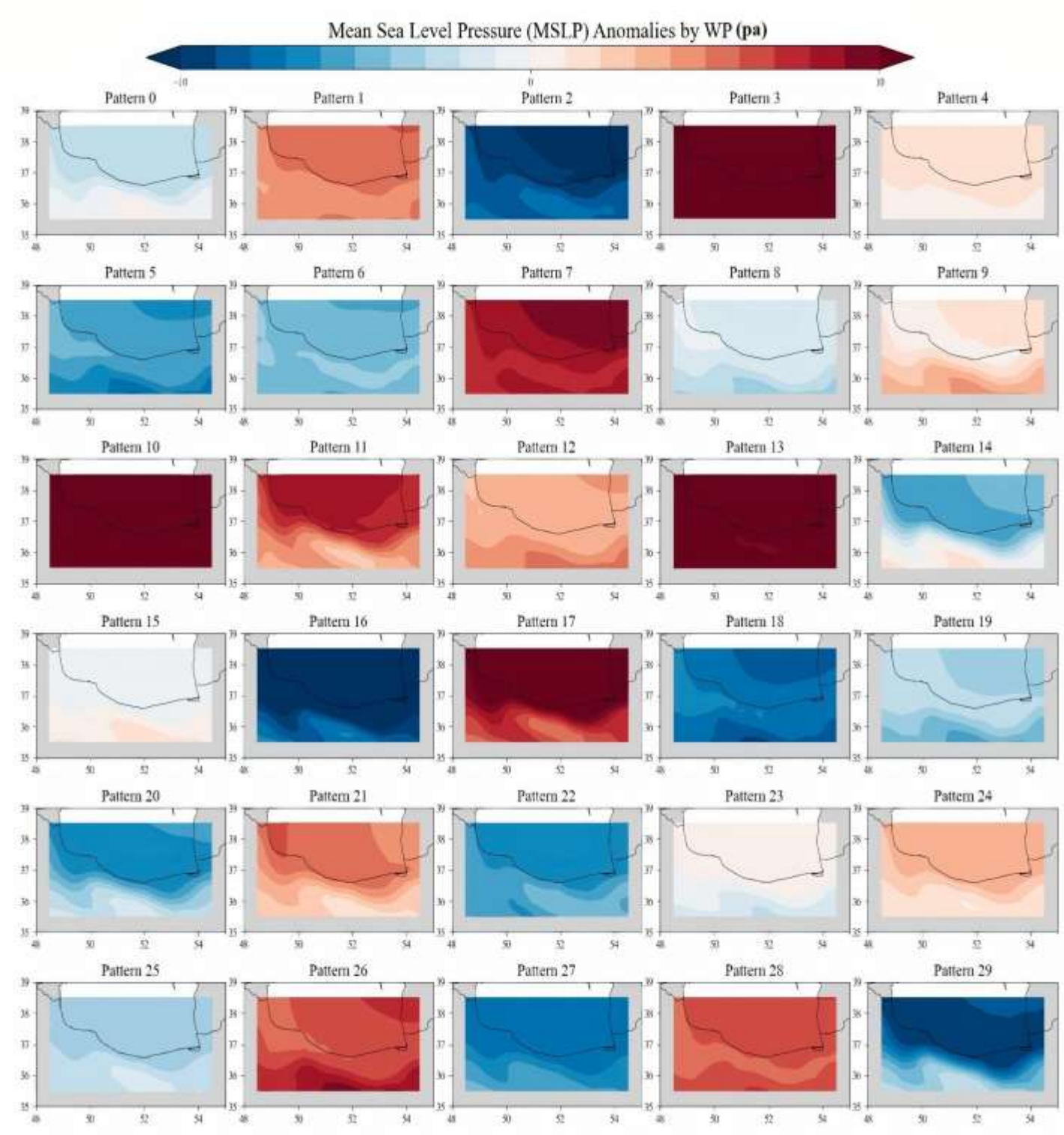


Fig 1. Map of sea surface pressure (MSLP) anomalies for 30 weather patterns (WPs) extracted from ERA5 reanalysis data during the tea growing season (April to October) in northern Iran during the period 1961–2023.

2. Materials and Methods

2.1. Study region

The main tea-producing regions of Iran are along the south coast of the Caspian Sea, such as the Gilan and Mazandaran provinces (Fig 2). The south of the Caspian Sea was the target of previous climate studies regarding sea surface temperature and precipitation variability [9], indicating that it is sensitive to atmospheric change. In addition, recent research suggested possible feedback between the volume of the Caspian Sea and large climate events, such as El Niño, via changes in jet stream position and regional regimes of precipitation [10]. These places have a humid subtropical climate, considerable rainfall, fertile alluvial soils, and mountainous relief—ideal conditions for tea farming. For the scope of this study, ERA5 climate data were collected for the domain between 35.5°N to 38.5°N latitude and 48.5°E to 54.5°E longitude, therefore encompassing the targeted tea-growing areas of northern Iran (Fig 2). The Alborz mountain range plays a central role in determining the climate of the area as it induces orographic precipitation and a mosaic of microclimates that have an important role in determining the growth patterns of tea. The ensuing interaction of elevation, temperature, and humidity—particularly in the tea-growing months of April to October—closely simulates the agroclimatic conditions of extensive tea-producing regions in South and Southeast Asia. To delimit the research area in the north of Iran, spatial data within the range 35.5 to 38.5 degrees north latitude and 48.5 to 54.5 degrees east longitude were used, including the main tea growing area in Gilan and Mazandaran provinces (Fig 2).

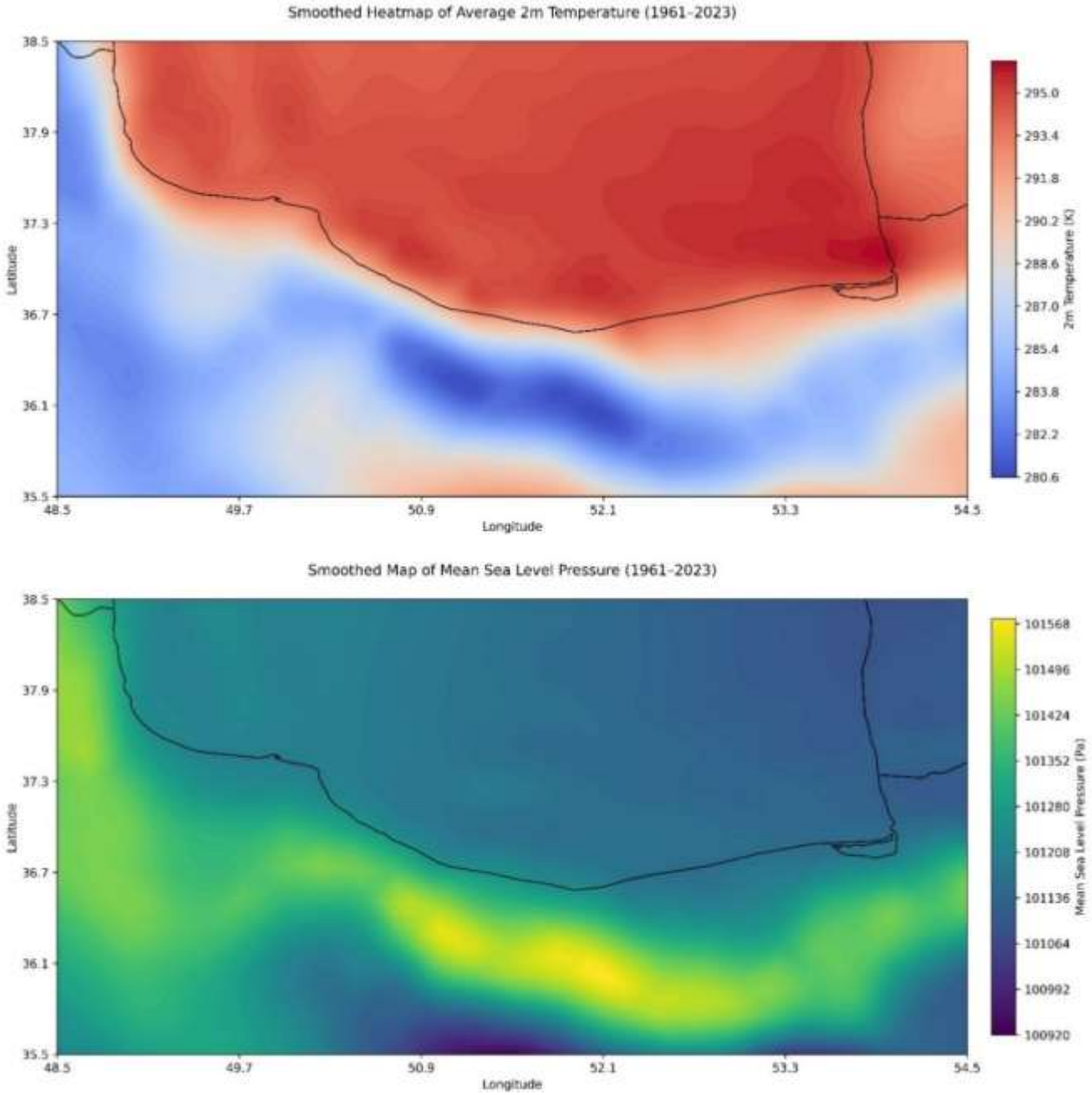


Fig 2. Overview of the study area in northern Iran, focusing on the central tea-producing provinces of Gilan and Mazandaran. The red and green color-based rectangles highlight the geographical domain used for climate data extraction from the ERA5, bounded by latitudes 35.5°N to 38.5°N and longitudes 48.5°E to 54.5°E.

2.2. Dataset

Annual tea yield data were sourced from the FAO (Food and Agriculture Organization of the United Nations) database . Since approximately 100% of Iran's tea production is concentrated in the provinces of Gilan and Mazandaran, the national-level data provided by FAO serve as a reliable proxy for regional yield estimation. The dataset includes annual records of total production

(in tons), harvested area (in hectares), and yield (in tons per hectare), spanning the period from 1961 to 2023[12].

Assessing the long-term impact of weather patterns (WPs) on tea yield presents significant analytical challenges, primarily due to the complex interplay of influencing factors across different temporal scales. These include short-term drivers such as intra-annual meteorological variability and long-term influences such as advancements in farming technologies, varietal improvements, agricultural policies, and the use of fertilizers and pesticides. As a result, tea yield data exhibit temporal trends and are statistically non-stationary—meaning their mean and variance fluctuate over time.

This study applied a simple detrending method based on the moving average technique to isolate inter-annual variability driven by climate and remove long-term trends associated with technological and structural changes in agriculture. Specifically, a moving average was computed using a five-year temporal window, and each year's yield was normalized relative to the corresponding window mean. The resulting series of values—yield anomalies—capture short-term fluctuations in tea yield attributable to inter-annual climatic variability. To isolate interannual variability, the yield series was detrended using a 5-year moving average window:

$$Y_{detrended} = Y_{observed} - MA(Y, 5)$$

where MA(Y, 5) denotes the 5-year moving average of the observed yield series[13].

The Spearman rank correlation coefficient was employed to assess the relationship between the frequency of specific weather patterns (WPs) and variations in tea yield. This non-parametric statistical measure is suitable for detecting linear and nonlinear associations in non-normally distributed datasets [14]. By correlating the annual frequency of each WP with the corresponding yield anomalies, this analysis identified key atmospheric configurations linked to yield variability, providing a statistically robust basis for understanding climate–agriculture interactions over the 63-year study period (1961–2023).

To investigate the influence of climate on inter-annual fluctuations in tea yield across northern Iran, this study utilized ERA5 reanalysis data provided by the Copernicus Climate Data Store [15]. ERA5 offers high-resolution spatial and temporal data and provides a comprehensive suite of essential climate variables, including 2-meter air temperature (T2M), mean sea level pressure (MSL), total precipitation (TP), surface solar radiation downwards (SSRD), and volumetric soil moisture across four depth layers (SWVL1 to SWVL4). The temporal scope of the analysis was limited to April through October for each year from 1961 to 2023, corresponding to the active tea growing season in the provinces of Gilan and Mazandaran.

For data preparation and preprocessing, daily records for each variable were extracted across a geographic domain encompassing the tea-producing regions of northern Iran. To reduce spatial

dimensionality and emphasize regional-scale dynamics, spatial averaging was applied across the entire study area for each variable. The resulting daily data were subsequently aggregated into yearly and annual means to align temporally with the tea yield dataset, which was only available annually. This procedure produced a comprehensive climate feature database suitable for multi-scale temporal analysis of tea–climate relationships, consistent with prior research where temporal data aggregation and machine learning models were applied to improve predictive performance in agricultural systems [17].

These data facilitated the examination of both intra-annual and inter-annual variability in key meteorological variables. When integrated with the annual frequency of weather pattern (WP) occurrences, they enabled the development of a comprehensive, multidimensional analytical framework for exploring the complex interactions between climate dynamics and tea yield. This integrative approach aligns with recent advances in climate–agriculture research and enhances the accuracy of assessments regarding the long-term impacts of climate variability and change on tea production. Moreover, by spanning over six decades of observational records, the framework provides a robust scientific basis for evidence-informed policymaking and the design of climate risk management strategies .

To assess the influence of Weather Patterns (WPs) on tea yield in northern Iran, it was first essential to extract these patterns based on the region's atmospheric conditions. For this purpose, daily Mean Sea Level Pressure (MSLP) data from the ERA5 reanalysis dataset—recognized for its high temporal resolution, global coverage, and data quality—were utilized . The extraction process involved several steps. MSLP data at midnight UTC were collected for the entire study period, spanning April 1961 to October 2023. To isolate atmospheric anomalies, the long-term daily climatological mean was calculated and subtracted from each day's observed MSLP value, yielding daily MSLP anomalies.

The MiniBatch KMeans clustering algorithm was employed to identify distinct synoptic-scale weather configurations. This algorithm, a computationally efficient variant of the traditional KMeans method, is well-suited for handling large-scale climate datasets and has been adopted in prior atmospheric classification studies such as the MO30 framework [18, 19]. In this study, multiple atmospheric variables—including mean sea level pressure (MSLP), geopotential height (Z), 2-meter air temperature (T2M), skin temperature (SKT), total precipitation (TP), and surface solar radiation downwards (SSRD)—were utilized as input features for the clustering. The clustering process partitioned the multidimensional anomaly fields into 30 distinct clusters, each representing a unique and persistent weather pattern (WP) characteristic of the northern Iran region. Following the identification of clusters, a unique code representing the dominant WP was assigned each day within the study period. Subsequently, each year, the number of days corresponding to each WP was tallied over the tea growing season (April to October), resulting in a frequency matrix used for subsequent analyses.

[number of years×30 patterns]

This WP frequency matrix served as the set of independent variables (features) for subsequent statistical and machine-learning analyses to model the relationship between atmospheric circulation patterns and inter-annual variations in tea yield. This integrated framework enabled a quantitative assessment of WP impacts on crop performance and laid the groundwork for developing predictive models and early warning systems to support climate-informed agricultural decision-making.

Cluster coherence was evaluated using the Calinski–Harabasz index, which yielded a high score (≈203,728), indicating that the resulting weather pattern clusters were well-separated and internally compact.

2.3 Data Processing and Statistical Analysis

2.3.1 Weather Patterns and Tea Yield Dynamics

Previous studies exploring the influence of Weather Patterns (WPs) on environmental systems have adopted mainly a meteorological lens, focusing on extreme events such as heatwaves, droughts, and intense precipitation episodes—typically emphasizing their statistical extremity and rarity [20]. In contrast, the present study takes an agriculture-centered perspective, assessing atmospheric patterns not solely through the lens of statistical intensity but rather by evaluating their direct agronomic impacts on tea growth and yield. This shift in focus facilitates a more nuanced understanding of the relationship between atmospheric variability and crop performance within the specific climatic context of northern Iran.

To support this analysis, annual tea yield data for 1961–2023 were obtained from the FAOSTAT database . Given that virtually all of Iran's tea production is concentrated in the provinces of Gilan and Mazandaran, the national-level statistics are considered a valid proxy for regional yield dynamics. Complementary climate data were drawn from the ERA5 reanalysis dataset, recognized for its high spatial and temporal resolution . These data were restricted to the tea growing season (April to October) and the geographic domain covering the primary tea-producing regions in northern Iran.

To evaluate the relationship between Weather Patterns (WPs) and tea yield, the annual frequency of daily occurrences for each identified WP (WP0 to WP29) was calculated. Rather than focusing on isolated extreme events, the analysis identified the WPs most commonly associated with high-, medium- and low-yield years.

The results revealed that specific patterns, notably WP1, WP18, and WP9, occurred significantly more often in high-yield years. These patterns were generally characterized by favorable agroclimatic conditions, including moderate temperatures, well-timed and evenly distributed precipitation, and sufficient solar radiation—all essential for optimal tea growth. Conversely, WP24, WP12, and WP0 patterns were more frequent during low-yield years and were

associated with adverse conditions such as elevated temperatures, reduced humidity, or extended dry spells [21].

In addition to correlation analysis, clustering and classification techniques were employed to validate and systematically distinguish between WPs with positive and negative impacts on yield. Building on these findings, a composite metric—termed the WP Index—was developed to quantify the annual difference between the frequencies of positive and negative WPs. The strong correlation between the WP Index and annual tea yield highlights its utility as a predictive indicator for distinguishing high- and low-yield years.

Overall, these findings underscore the potential of WPs as powerful diagnostic tools for characterizing climate variability and its effects on agricultural productivity. The integration of WP classification with yield data offers a robust, evidence-based framework for informing resource allocation, optimizing crop management, and strengthening the climate resilience of agricultural systems in northern Iran [22, 23].

2.3.2 Climate Change Impact Assessment

This study utilized long-term historical data on the frequency of Weather Patterns (WPs) from 1961 to 2023 to examine their relationship with annual tea yield in Iran's northern provinces of Gilan and Mazandaran. Specifically, the analysis focused on the tea growing season (April to October), during which the influence of atmospheric circulation is most relevant to crop development. The strength of this approach lies in its use of synoptic-scale atmospheric configurations rather than isolated climate variables (e.g., temperature or precipitation), thereby offering a more integrated representation of the meteorological conditions shaping plant growth.

Due to the lack of region-specific future projections for WP frequency in Iran, the study adopted a scenario-based methodology to explore the potential impacts of climate change under plausible future conditions [24]. Within this framework, the previously identified WPs—derived via clustering algorithms—were classified into two categories: Positive WPs, which were more frequently observed in high-yield years, and Negative WPs, which tended to dominate in years with reduced yield.

To quantitatively evaluate the balance between favorable and unfavorable atmospheric conditions in a given year, a composite metric termed the WP Index was introduced, defined as follows:

$$\text{WP Index} = \frac{N^{+} - N^{-}}{N^{+} + N^{-}}$$

Where $N+$ represents the number of days with positive WPs and $N-$ with negatives, in this formulation, the WP Index ranges from –1 to +1, where positive values denote a predominance of favorable climatic conditions. In contrast, negative values indicate years characterized by a

dominance of adverse weather patterns. After computing the index for each year in the study period, its association with annual tea yield was analyzed. The results revealed a clear pattern: years with higher WP Index values corresponded to higher tea yields, while lower values were associated with reduced productivity. These findings underscore the utility of the WP Index as a straightforward yet effective tool for assessing interannual atmospheric variability and forecasting crop performance [25].

Two hypothetical scenarios were constructed to explore the potential consequences of climate change further. In the first scenario, the frequency of positive WPs increases; in the second, negative WPs become more prevalent. When compared with yield records, historical trends in the WP Index suggest that this index can serve as a valuable indicator for assessing the potential impacts of climate change on tea production within a scenario-based framework.

Should future projections of WP frequency—derived from regional or global climate modeling efforts such as CORDEX and CMIP6—become available, this framework's predictive precision and spatial granularity could be further enhanced. Such advancements would allow for more robust regional-scale scenario analyses and support the development of targeted, evidence-based adaptation strategies to improve the climate resilience of agricultural systems[22, 24].

2.3.3 Long-Term WP-Yield Relationships

Unlike many previous studies that have primarily relied on qualitative assessments or focused on specific years, the present research adopts a comprehensive, quantitative approach based on long-term data from 1961 to 2023. Within this framework, the daily frequency of each Weather Pattern (WP) during the key months of the tea growing season (April to October) was extracted and integrated with annual tea yield records obtained from the FAOSTAT database . This methodology ensured that the analyses were grounded in reliable, empirical data and reduced the influence of random interannual fluctuations.

Spearman's rank correlation coefficient was first employed to assess the quantitative relationship between WPs and tea yield. As a non-parametric statistical tool, this method is well-suited for identifying linear and nonlinear associations in non-normally distributed datasets . To further enhance the precision of yield modeling and capture more complex climate-yield interactions, a suite of machine learning algorithms was implemented, including Support Vector Regression (SVR), Decision Trees, Random Forests, Logistic Regression, and the k-nearest Neighbors (KNN) algorithm [26]. These models enabled the identification of intricate, interactive patterns between atmospheric conditions and tea yield dynamics.

These analyses indicated that specific weather patterns—such as WP1 and WP18—were consistently associated with high-yield years, whereas patterns like WP24 were predominantly

observed during low-yield years. These findings underscore the pivotal role that shifts in WP frequency play as a primary driver of inter-annual tea yield variability in northern Iran. A key strength of this study is its long-term perspective, which allowed for identifying cumulative and persistent effects of atmospheric circulation patterns. Unlike short-term variability that may reflect random fluctuations, several weather patterns demonstrated sustained correlations with yield over multiple decades and were thus identified as key determinants of tea productivity. Ultimately, the integrated analytical framework employed in this study—combining traditional statistical methods with modern machine learning algorithms—offers a robust tool for gaining deeper insights into the complex interactions between climate and agriculture [29]. This approach provides a strong foundation for developing decision-support tools, evidence-based adaptation policies, and strategies to enhance agricultural systems' climate resilience [23].

To facilitate a clearer understanding of the research methodology, a flowchart has been provided to summarize the key steps of the study visually. This diagram offers a simplified overview of the process, helping readers grasp the sequence of actions, their interconnections, and the underlying logic of the methodological framework (Fig 3).

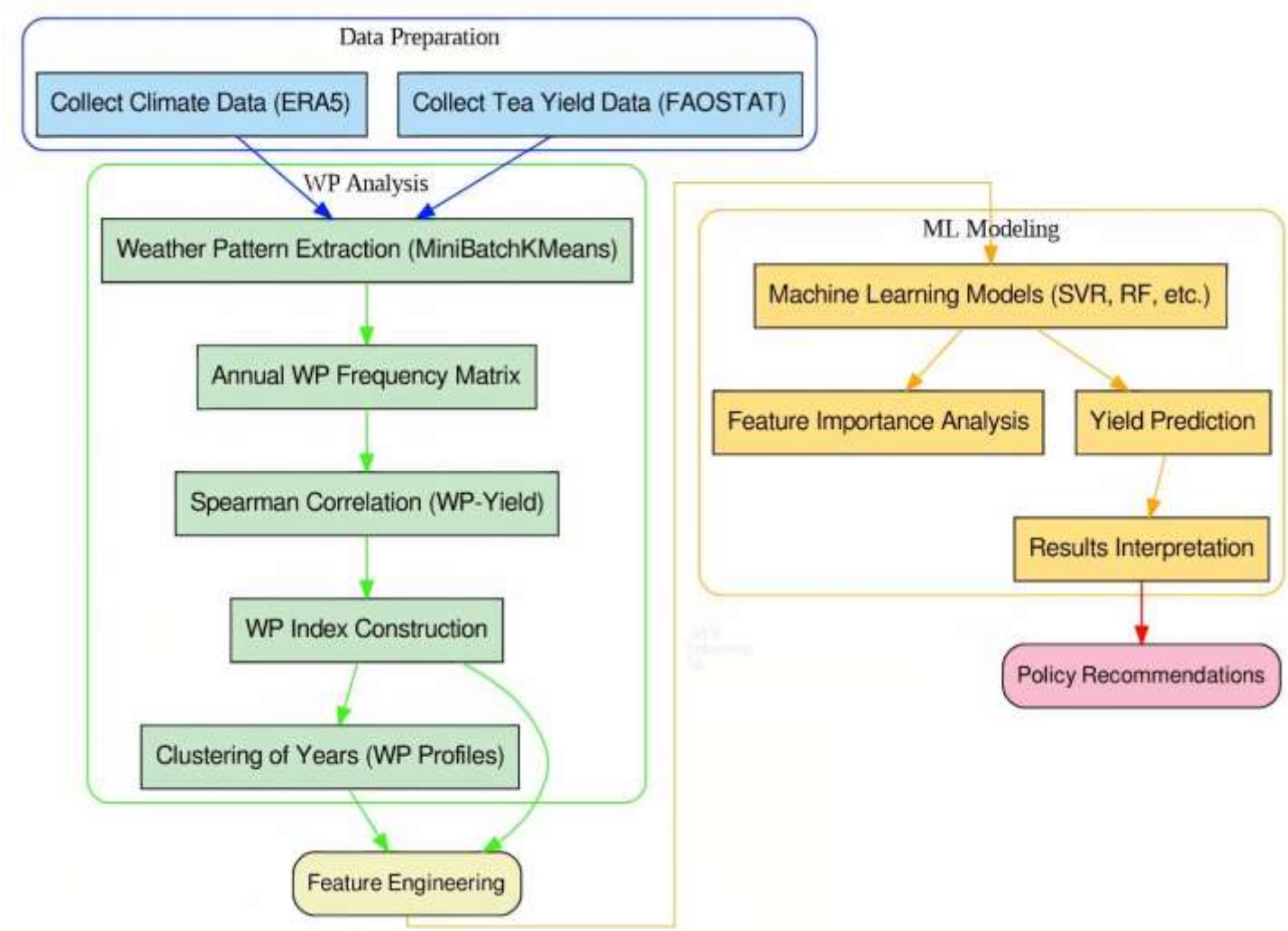


Fig 3. Schematic representation of the methodological framework adopted in this study to investigate the relationship between climate-derived weather patterns and tea yield variability in northern Iran.

Results and discussion

3.1 Distribution of Weather Patterns During the Tea Growing Season

To better understand the dominant climatic behaviors during the tea growing season, the frequency distribution of Weather Patterns (WPs) was analyzed for the key phenological months—April to October—from 1961 to 2023 in Iran's Gilan and Mazandaran provinces. This analysis was based on daily data derived from the WP classification process, where each day in the study period was assigned one of 30 identified atmospheric patterns .

The results revealed a markedly uneven and asymmetric distribution of WP occurrences over the study period. In other words, specific patterns contributed disproportionately to the formation of

prevailing atmospheric conditions in northern Iran. WP18, WP4, WP3, and WP19 emerged as the most frequent, making them the dominant synoptic regimes during the tea growing season. These patterns were generally associated with meteorological features such as regular rainfall or short-duration dry spells—conditions that play a critical role in determining soil moisture levels, air temperature, and crop performance. In contrast, patterns such as WP6, WP29, and WP21 occurred far less frequently, indicating a more limited role in shaping the regional climate. Nonetheless, the importance of these rare patterns should not be underestimated, as they may trigger extreme events—such as severe droughts or flash floods—with potentially adverse impacts on crop productivity under specific conditions. Similar observations have been reported in Europe, where rare but intense weather events—like the extreme heat and drought of 2018—led to significant yield losses in multiple crop types .

Moreover, this heterogeneous distribution of WPs offers a plausible explanation for the observed inter-annual variability in tea yield. For instance, frequent precipitation-driven patterns may enhance soil moisture availability, leading to improved leaf development and higher yield. Conversely, the dominance of hot and dry patterns—particularly during critical stages such as the first and second harvests—can result in significant yield reductions [27, 28].

This WP distribution also provides the foundation for more advanced analyses, such as exploring differences in atmospheric behavior between high- and low-yield years and developing composite climatic indices. A better understanding of these patterns' temporal frequency and distribution enables more accurate yield forecasting and climate risk management. Ultimately, the insights from this section establish the groundwork for identifying the most influential WPs in subsequent stages of the study and for analyzing their impacts both quantitatively and qualitatively.

3.2 Classification of High- and Low-Yield Years

To assess the relationship between weather patterns and tea yield, the first step involved examining annual tea yield data for Iran from 1961 to 2023. These data, obtained from FAO reports, reflect national-level yield figures, which—given the geographic concentration of tea production in the provinces of Gilan and Mazandaran—can be considered a valid proxy for regional yield performance in northern Iran . For the classification of years, the metric "tea yield per hectare" (ton/ha) was used . Furthermore, to identify the most influential synoptic patterns, the Pearson correlation coefficient was calculated between the annual frequency of each WP and tea yield. The correlation between the yearly frequency of individual WPs and tea yield was then explored using pairwise scatter plots with linear trend lines to identify potentially influential patterns (Fig 4).